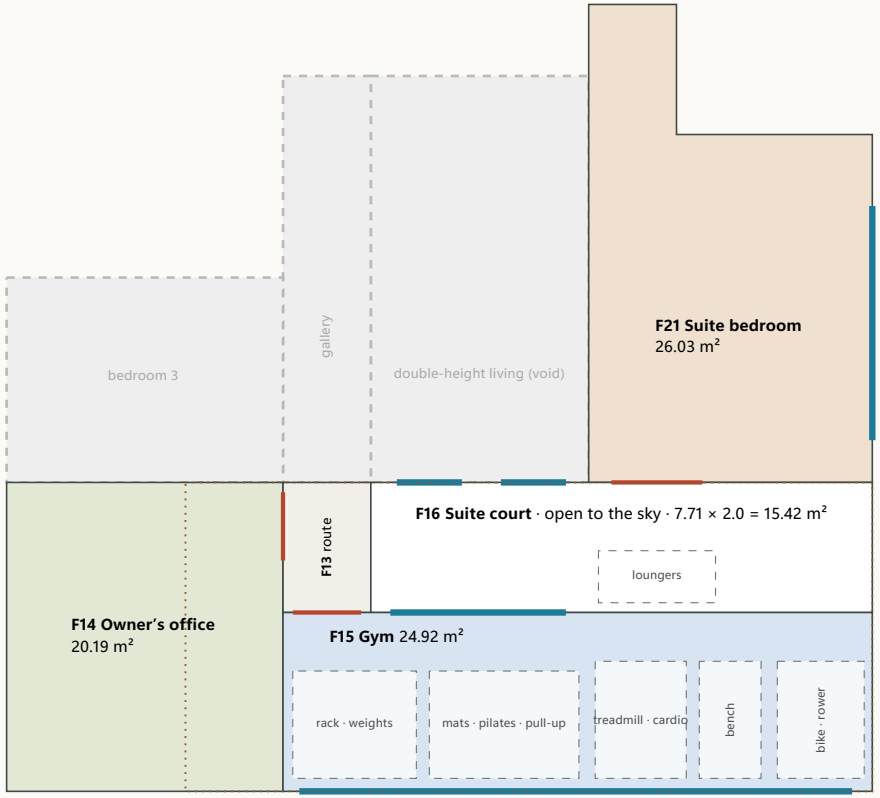
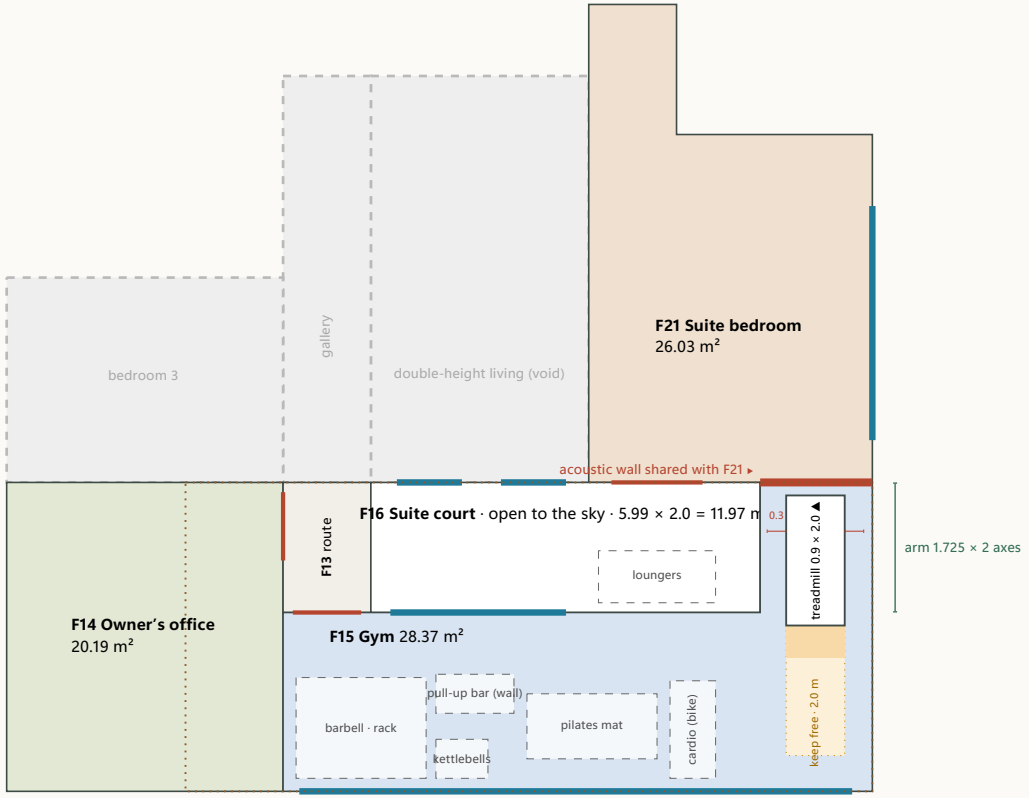
First floor · gym variant study · an L-shaped gym

Taif / Palm House · on V5.0 (tag v5.0) · 24 Oct 2026 · branch v5-gym-L · main files unchanged · CSV coordinates, wall centre lines, north up



V5.0 · gym a rectangle



Variant · L-shaped gym, treadmill arm

Red = doors (all sliding here) and, right, the shared wall arm / suite bedroom · blue = windows (the two clerestory windows of the double-height living on y 19.25, the gym's high windows onto the court on y 21.25) · orange = the treadmill's rear zones (dark: the owner's 0.5 m; light dotted: 2.0 m, kept free) · brown dotted = the GF men's majlis below (no inner wall).

1 · V5.0 geometry (data/rooms_ff.csv = studies/ff_suite.typ)

Room	Corners (x, y)	Axes	m²
F15 Gym	x 4.25–13.311, y 21.25–24.0	9.061 × 2.75	24.92
F13 Gym route	x 4.25–5.6, y 19.25–21.25	1.35 × 2.0	2.7
F14 Owner's office	x 0–4.25, y 19.25–24.0	4.25 × 4.75	20.19
F16 Suite court	x 5.6–13.311, y 19.25–21.25	7.711 × 2.0	15.42
F21 Suite bedroom	polygon to x 13.311, y 19.25	4.361 × 7.35 (L)	26.03

Shared walls: gym / office x 4.25 (y 21.25–24) · gym / gym route y 21.25 (x 4.25–5.6, gym door) · gym / suite court y 21.25 (x 5.6–13.311, high windows) · route / office x 4.25 (office door) · route / court x 5.6 · court / suite bedroom y 19.25 (x 8.95–13.311, the bedroom's sliding door) · court / double-height living y 19.25 (x 5.6–8.95, the two clerestory windows).
Answer: yes, the gym is a plain rectangle 9.061 × 2.75 m (24.92 m²; it was an L on 18 Oct and was made a rectangle in V5.0 on 20 Oct). It does **not** touch the suite bedroom: the 2.0 m suite court lies between them.

2 · The short arm, computed

- **Width:** treadmill 0.9 + 2 × 0.3 = **1.5 m clear** → + ½ outer wall (0.125, east) + ½ wall (0.10, to the court) = **1.725 m on axes** (x 11.586–13.311).
- **Length:** the arm can only run north up to the suite bedroom's south wall (y 19.25): **2 m on axes** (y 19.25–21.25, the court's depth), 1.9 m clear inside it. The treadmill needs 2.6 m clear from the bedroom wall face (0.1 gap + 2.0 + 0.5 behind): it stands in the arm and **steps off south into the long arm**, where the 0.5 m (and the 2.0 m of the safety standards) lies. A 2.5 m arm on axes would enter the bedroom.
- **Treadmill:** x 11.986–12.886, y 19.45–21.45, facing north; 0.3 m each side, 0.1 m off the bedroom wall (no contact).
- **Gym:** long arm 9.061 × 2.75 = 24.92 m² + short arm 1.725 × 2 = 3.45 m² → **28.37 m²** (V5.0 24.92).

✓ Treadmill clearances (0.3 each side, 0.5 and 2.0 m behind) free · furniture inside the gym, not overlapping, out of the treadmill zones and the gym door · bedroom door and clerestory windows still on the court · areas add up

Gym variant · rectangle (V5.0) against the L

Comparison

Item	V5.0 · rectangle	Variant · L
Gym (F15)	9.061 × 2.75 = 24.92 m²	28.37 m² (long 24.92 + arm 3.45)
Long arm	–	9.061 × 2.75 m, as now
Short arm (treadmill only)	–	1.725 × 2 m axes · 1.5 × 1.9 m clear
Treadmill	in the row of machines along the south wall (treadmill · cardio)	in the arm, facing north; 0.3 m each side; 2.42 m free behind (≥ 2.0 kept)
Long-arm equipment	rack · mats/pilates/pull-up · treadmill/cardio · bench · bike/ rower	barbell + rack · kettlebells · pull-up bar · pilates mat · one cardio (bike); no bench
Suite court (F16)	7.711 × 2.0 = 15.42 m²	5.99 × 2.0 = 11.97 m² (−3.45)
Court open to the sky, private	yes · walled on 4 sides	yes · walled on 4 sides (east side now the arm's solid wall)
Clerestory windows: winter-noon sun	81 %	81 % (unchanged)
Clerestory windows: equinox-noon sun	100 %	100 % (unchanged)
Clerestory windows: sky seen from the living	100 %	100 % (unchanged)
Gym touches the suite bedroom	no (court between)	**yes** · wall y 19.25, x 11.586–13.311 (1.725 m)
Suite bedroom door to the court	x 9.3–10.7, sliding	unchanged (0.79 m to the arm's wall)
Suite bedroom windows	east x 13.311, y 15.0–18.6	unchanged
New walls on the men's majlis roof	y 21.25 (x 5.6–13.311)	y 21.25 (x 5.6–11.586) + arm west wall x 11.586 (y 19.25–21.25) + arm roof
Roofed FF area	–	+3.45 m² (court → gym)

Light: the section method of design_ff.typ page 4 (sill +5.00, head +6.00, the gym's north wall a railing with solid top +7.20, 2.0 m in front of the windows, seated eye +1.20 at y 16.1). The arm stands 2.99 m east of the east clerestory window; with true north 21.4° east of the M line the noon sun stands south-south-west of the plan frame, so the arm's shadow falls north-east, away from the windows (it can shade them only on early mornings).

The suite court after the change

- 5.99 × 2.0 m = 11.97 m² (was 15.42): still **open to the sky** and **private**, walled on four sides: north the double-height living (clerestory) and the suite bedroom (sliding door), south the gym's north wall (high obscured windows), west the gym route, **east now the treadmill arm's solid wall** (no window onto the court, or high obscured glass).
- The two clerestory windows (x 6.0–7.0 and 7.6–8.6) keep the full 2.0 m of open court in front of them: winter-noon sun 81 %, sky 100 %, as in V5.0.
- **Rain:** V5.0 lets the court fall to the east; that end is now the arm. Move the floor drain to the court's new east end (x ≈ 11.19) or fall it west; the arm's roof drains outside like the gym roof.

Acoustics · the wall between the treadmill and the suite bedroom

- **The problem** is mostly **structure-borne:** footfall impact and motor vibration go into the slab and the wall, then into the bedroom. Airborne noise comes second.
- **Wall** (y 19.25, x 11.586–13.311, and its junctions with the outer wall and the arm wall): a **double-leaf wall** rather than one 200 mm block: e.g. the 200 mm dense block + a free-standing stud lining on the gym side (no contact, 50 mm mineral wool, 2 × 12.5 mm acoustic plasterboard), or two block leaves with a 50 mm filled cavity. Target ≈ **Rw ≥ 55 dB** (DnT,w ≥ 50 dB between gym and bedroom); seal the top, no sockets back to back, no duct through it.
- **Floor under the arm:** a **floating floor** (resilient layer + 50–70 mm screed, perimeter isolation strip, not touching the walls) or at least 20–40 mm dense rubber gym tiles over the whole arm, plus **anti-vibration pads** under the treadmill feet. Stop the floating screed short of the bedroom wall.
- **Treadmill:** 0.1 m off the shared wall, never fixed to it; choose a model with a cushioned deck and a quiet (AC) motor.
- **Use:** the bedroom is the master bedroom; a house rule (no running while someone sleeps) costs nothing.

Structure (men's majlis roof, no wall below)

- The GF men's majlis (x 2.75–13.311, y 19.25–24) has **no inner wall**. On its roof the variant keeps the gym's north wall y 21.25 (now x 5.6–11.586) and adds the **arm's west wall x 11.586, y 19.25–21.25**, with no wall below. Its north wall (y 19.25) is on the GF living / men's majlis wall and its east wall is the outer wall.
- **New:** the arm's roof slab (3.45 m²) and the treadmill's dynamic load (≈ 1.5–2 kN in use) on the majlis roof slab. Engineer: a light wall or a beam on x 11.586, the slab for the treadmill's vibration, and the floating floor's weight.

Treadmill safety

- The owner's minimum (0.5 m behind, 0.3 m each side) is met. The equipment standards (ASTM F2115, EN 957-6) ask for **2.0 × 0.9 m free behind and 0.5 m each side**: the 2.0 m behind is kept free in the long arm (drawn); for 0.5 m sides the arm would be 2.125 m on axes (court 5.59 m long).

Questions for the owner

1. Accept the smaller suite court (11.97 m², was 15.42)?
2. The bench is dropped from the long arm (not on the list); keep one?
3. Arm sides 0.3 m (owner) or 0.5 m (standard)?